Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	*(a)*			Substitution: $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{12} + \frac{1}{6}$ **or** $R = \frac{12 \times 6}{12+6}$ (1) R = 4 [Ω] (1) $R = R_1 + R_2$ = 4 (**ecf** only from use of parallel equation) + 16 = 20 [Ω] (1) Answer of 16.25 [Ω] award 2 marks	1	1 1		3	3	3
	(b)			$P = I^2R$ = 1.5^2 × 20(**ecf**) = 45 [W] (1) Substitution: $t = \frac{75.6\,[\text{x}\,10^3]}{45(\textbf{ecf})}$ (1) = 1 680 [s] (1) $\frac{1680(\textbf{ecf})}{60}$ = 28 [minutes] (1) Answer of 1.68 × 10^n where n is not equal to 3 award 2 marks Answer of 2.8 × 10^n where n is not equal to 1 award 3 marks	1	1 1 1		4	4	4
				Question 5 total	**2**	**5**	**0**	**7**	**7**	**7**